

MCQs: Database Concepts (Class XII – Computer Science)

INTRODUCTION & NEED OF DATA MANAGEMENT

1. The primary purpose of storing data electronically is to:
 - A) Increase paperwork
 - B) Improve manual calculations
 - C) Store and manage data efficiently
 - D) Reduce number of users
2. In the school attendance example, what is recorded daily by the class teacher?
 - A) Marks
 - B) Attendance status
 - C) Guardian details
 - D) Student address
3. Attendance status marked by the teacher includes:
 - A) Y and N
 - B) 1 and 0
 - C) P and A
 - D) Yes and No
4. If class strength is 50 and working days are 26, how many attendance records are created per month?
 - A) 130
 - B) 260
 - C) 1300
 - D) 2600
5. Manual data entry becomes tedious mainly due to:
 - A) Low storage capacity
 - B) Increased data volume
 - C) Lack of electricity
 - D) High hardware cost

LIMITATIONS OF MANUAL RECORD KEEPING

6. Rewriting student details during promotion leads to:
 - A) Data normalization
 - B) Data redundancy
 - C) Data encryption
 - D) Data indexing
7. Incorrectly written names and skipped records cause:
 - A) Data isolation
 - B) Data inconsistency
 - C) Data security
 - D) Data abstraction

8. Loss of attendance register results in:
- A) Data dependency
 - B) Data redundancy
 - C) Permanent data loss
 - D) Faster processing
9. Errors during manual consolidation mainly affect:
- A) Data storage
 - B) Data accuracy
 - C) Data encryption
 - D) Data abstraction
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FILE SYSTEM CONCEPT

10. A file in a computer system is best described as:
- A) A database
 - B) A container to store data
 - C) A query
 - D) A schema
11. Which of the following can be file content?
- A) Text only
 - B) Program code only
 - C) Images and videos only
 - D) All of the above
12. CSV stands for:
- A) Computer Stored Values
 - B) Common Separated Values
 - C) Comma Separated Values
 - D) Column Stored Values
13. To display attendance on a website using files, one must:
- A) Use DBMS
 - B) Write application programs
 - C) Use spreadsheets
 - D) Use hardware tools
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STUDENT & ATTENDANCE FILE STRUCTURE

14. The STUDENT file contains how many columns?
- A) 4
 - B) 5
 - C) 6
 - D) 7
15. Which attribute stores guardian phone number?
- A) GName
 - B) GAddress

- C) GPhone
 - D) SDateofBirth
16. Which file is maintained by the class teacher?
- A) STUDENT
 - B) GUARDIAN
 - C) ATTENDANCE
 - D) DATABASE
17. AttendanceStatus can have values:
- A) Yes / No
 - B) True / False
 - C) P / A
 - D) 1 / 0
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LIMITATIONS OF FILE SYSTEM

18. Difficulty in accessing data arises because:
- A) Files are encrypted
 - B) Files do not provide retrieval mechanisms
 - C) Files are read-only
 - D) Files are too large
19. Data accessed through file systems requires:
- A) SQL queries
 - B) Application programs
 - C) DB engines
 - D) Metadata
20. Repetition of student names in multiple files is an example of:
- A) Data inconsistency
 - B) Data isolation
 - C) Data redundancy
 - D) Data dependency
21. Data redundancy leads to:
- A) Faster processing
 - B) Less storage
 - C) Data inconsistency
 - D) Data security
22. When same data differs in multiple files, it is called:
- A) Data abstraction
 - B) Data inconsistency
 - C) Data isolation
 - D) Data integrity
23. Lack of linking between files causes:
- A) Data redundancy
 - B) Data isolation

- C) Data inconsistency
 - D) Data dependence
24. Change in file structure affecting programs is known as:
- A) Data independence
 - B) Data dependency
 - C) Data isolation
 - D) Data abstraction
25. Providing read-only access is difficult in file systems due to:
- A) Data redundancy
 - B) Lack of access control
 - C) Large file size
 - D) Lack of storage

DATABASE MANAGEMENT SYSTEM (DBMS)

26. DBMS stands for:
- A) Data Backup Management System
 - B) Database Management System
 - C) Data Block Management System
 - D) Database Modeling Software
27. DBMS is mainly used to:
- A) Write programs
 - B) Create and manage databases
 - C) Design hardware
 - D) Create files
28. Which of the following is NOT a DBMS?
- A) MySQL
 - B) Oracle
 - C) PostgreSQL
 - D) HTML
29. DBMS provides users with:
- A) Physical data view
 - B) Abstract data view
 - C) Manual data view
 - D) Printed data view
30. Retrieving data using commands is called:
- A) Compilation
 - B) Querying
 - C) Debugging
 - D) Formatting

DATABASE APPLICATIONS

31. Banking databases maintain:
- A) Student attendance
 - B) Loan and transaction details
 - C) Inventory stock
 - D) Sports records
32. Inventory management databases store:
- A) Exam results
 - B) Employee salary
 - C) Product and order details
 - D) Crop data
33. Online shopping databases store:
- A) Employee data
 - B) User preferences
 - C) Attendance data
 - D) Examination records
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FILE SYSTEM TO DBMS TRANSITION

34. Tables in a database are linked using:
- A) Indexes
 - B) Queries
 - C) Common fields
 - D) Programs
35. RollNumber is common between STUDENT and:
- A) GUARDIAN
 - B) DATABASE
 - C) ATTENDANCE
 - D) SCHEMA
36. Removing SName from ATTENDANCE helps reduce:
- A) Dependency
 - B) Redundancy
 - C) Isolation
 - D) Vulnerability
37. Splitting STUDENT into STUDENT and GUARDIAN avoids:
- A) Data inconsistency
 - B) Data redundancy
 - C) Data loss
 - D) Data encryption
38. GUID is created to:
- A) Store phone numbers
 - B) Identify guardians uniquely
 - C) Store student marks
 - D) Encrypt data

39. Phone number is not ideal as guardian identifier because:

- A) It is numeric
- B) It may change
- C) It is long
- D) It is private

DATABASE SCHEMA & CONCEPTS

40. Database schema represents:

- A) Stored data
- B) Database design
- C) Database instance
- D) Query result

41. Schema includes:

- A) Only table names
- B) Only constraints
- C) Structure and relationships
- D) Only data

42. Constraints ensure:

- A) Data speed
- B) Data accuracy
- C) Data encryption
- D) Data storage

43. Meta-data is:

- A) Actual data
- B) Data about data
- C) Query output
- D) Backup data

44. Database instance refers to:

- A) Empty database
- B) Database structure
- C) Snapshot at a given time
- D) Database software

QUERY & DATA MANIPULATION

45. Query is a request for:

- A) Database deletion
- B) Information retrieval
- C) Hardware access
- D) File creation

46. Insertion operation means:

- A) Modifying data
- B) Adding new data

- C) Removing data
- D) Sorting data

47. Updating guardian phone number is an example of:

- A) Deletion
- B) Insertion
- C) Update
- D) Query

48. Removing student records is called:

- A) Insertion
- B) Update
- C) Deletion
- D) Retrieval

DATABASE ENGINE

49. Database engine is responsible for:

- A) Data entry only
- B) Query handling
- C) Hardware control
- D) User interface only

RELATIONAL DATA MODEL

50. The most commonly used data model is:

- A) Hierarchical
- B) Object-oriented
- C) Relational
- D) Document

51. Tables in relational model are called:

- A) Files
- B) Schemas
- C) Relations
- D) Records

52. Each row in a table represents:

- A) Attribute
- B) Domain
- C) Tuple
- D) Schema

53. Columns in a relation are called:

- A) Tuples
- B) Attributes
- C) Domains
- D) Instances

RELATION TERMINOLOGY

54. Domain of an attribute represents:
- A) Table size
 - B) Set of possible values
 - C) Number of rows
 - D) Attribute name
55. Degree of a relation means:
- A) Number of rows
 - B) Number of keys
 - C) Number of attributes
 - D) Number of databases
56. Cardinality of a relation means:
- A) Number of columns
 - B) Number of attributes
 - C) Number of tuples
 - D) Number of keys

PROPERTIES OF A RELATION

57. Attribute names in a relation must be:
- A) Same
 - B) Unique
 - C) Numeric
 - D) Optional
58. Order of attributes in a relation is:
- A) Fixed
 - B) Important
 - C) Immaterial
 - D) Mandatory
59. Tuples in a relation must be:
- A) Ordered
 - B) Identical
 - C) Distinct
 - D) Null
60. Atomic value means:
- A) Multiple values
 - B) Divisible value
 - C) Indivisible value
 - D) Calculated value
61. NULL represents:
- A) Zero
 - B) Empty string

- C) Unknown or not applicable
 - D) False value
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KEYS IN RELATIONAL DATABASE

- 62. Candidate key is an attribute that:
 - A) Can be NULL
 - B) Is always foreign
 - C) Uniquely identifies tuples
 - D) Is always composite
- 63. Primary key is selected from:
 - A) Foreign keys
 - B) Candidate keys
 - C) Alternate keys
 - D) Composite keys
- 64. Alternate keys are:
 - A) Primary keys
 - B) Remaining candidate keys
 - C) Foreign keys
 - D) Composite keys
- 65. Composite primary key consists of:
 - A) One attribute
 - B) Two or more attributes
 - C) Only foreign keys
 - D) Only numeric keys
- 66. Foreign key is used to:
 - A) Store data
 - B) Encrypt data
 - C) Relate tables
 - D) Delete records
- 67. Foreign key value is derived from:
 - A) Candidate key
 - B) Primary key of another table
 - C) Alternate key
 - D) Domain
- 68. Foreign key can have NULL values because:
 - A) It is always optional
 - B) It may not be part of primary key
 - C) It stores text
 - D) It stores integers